

COINPICKS · RESEARCH & EDUCATION

Bitcoin Technical Analysis Quant Pillar Report

The Tape Itself — Measured, Scored, and Mapped to the Pillar Standard

*Machine-computed from 16 years of daily price data (5,857 days, 2010–2026) through a 7-indicator scoring stack, halving-cycle analogs, and a power-law valuation model. Every number is reproducible from the scripts in the BTC-TA folder. **Scores are the model's draft — awaiting my stamp.***

COINPICKS QUANT PILLAR REPORT · THE QUANTITATIVE SLEEVE OF MY THESIS

Data through July 30, 2026 (BTC \$64,706) · Pipeline re-run & verified July 30, 2026 (full weekly refresh) · Not Financial Advice

The Scores — July 30 Model Draft

Model draft computed July 30, 2026 (full weekly refresh; previous run July 25) — same standard as every qualitative pillar. Adjust any P and re-stamp. (July 25 draft was 40/36/43/68/70 — the tape eased at the front end this cycle while the cycle-analog windows firmed slightly.)

WINDOW	DRAFT P	QUALITY	IMPACT	SIGNAL	CALL
1 month	39	0.75	8	-1.32	Mild Bear — broken tape, repair signal #1 (200-week MA) still holding
3 months	37	0.70	8	-1.46	Mild Bear — the final-flush window, 0-of-3 analog
6 months	46	0.55	8	-0.35	Neutral (slight bear lean)
1 year	67	0.60	8	+1.63	Bull — the cycle turn
3 years	70	0.50	8	+1.60	Bull — value + cycle

Impact: 8 (the tape aggregates every driver, but adds no causal force of its own) · **Formula:** Signal = ((P – 50) ÷ 50) × Quality × Impact · **Honesty cap:** P = 50 + 25 × (window score ÷ 100) — pure TA never claims odds more than 25 points from a coin flip, because its measured out-of-sample edge is modest. Math machine-verified.

THE THREE INPUTS, IN PLAIN ENGLISH

- P** — the odds the tape is *good for Bitcoin* by then. Near-term low (weekly composite –50/100, all seven indicators still bearish — though price is holding above the 200-week average); long-term high (at this exact point in prior cycles, the next 12 months were positive 3-for-3, and price sits 53% below its power-law fair value).
- Quality** — how trustworthy the evidence is. Highest near-term (835 weekly bars, unanimous indicators), lowest at 3 years (only 3 completed cycles to learn from, and this cycle is already breaking the old pattern).
- Impact** — 8 out of 10, now the uniform value shared by every pillar; the TA pillar carries an equal 20% weight in the master, same as each qualitative pillar. Price is the thing itself — but a mirror, not an engine.

The Claim

The thesis, and where the numbers differ from the mood

- The crowd:** –48% from the all-time high, a lower low on June 30, a weekly close under the 200-week average as recently as June 28 — "the cycle is dead, get out." (Price closed at \$64,706 (+10.6% off the June 30 low) through a week in which the Iran war widened to two new countries and the Fed delivered its most hawkish FOMC dissent in a decade — both events the qualitative pillars treat as materially bearish. That resilience is the counter-evidence the crowd is ignoring.)
- The model:** the short-term tape is still broken (P 39/37 for 1–3 months — this is not yet a dip-buying report), and the 6-month window is still a near coin flip with a slight bear lean (P 46). But the same data says P 67 at one year: every prior cycle at this exact month looked just as dead and was higher 12 months later, price is deep below fair value, and the froth that precedes real crashes is absent.
- The edge:** the crowd reads one clock (the tape). The model reads three — tape (bearish), cycle clock (bottoming window: prior cycles bottomed at months 26–30; we're at month 27.33), and valuation (53% below the power-law fair line, under the –1σ "cheap" band). Two of the three clocks are already on the bull's side; the tape is the last one, and it always turns latest.

The Facts — The Tape

Computed July 30, 2026 from Bitstamp daily data (BTC \$64,706, daily close as of July 30). Current weekly/monthly/quarterly bars are still forming — scores use them as-is. Quantified: on completed bars only, the weekly composite is -29.2, monthly -42.6, quarterly +7.7.

- **Weekly composite -28.3/100 (Slight Bearish), 6-of-7 indicators agree.** Price is below the 20-week (\$69,428, -6.8%) and 50-week (\$84,371, -23.3%) averages but holding **above the 200-week (\$63,557, +1.8%)**; RSI 40.0; MACD line below zero (-5,183) but histogram turned positive (+327) — early momentum-repair signal; %B 0.32.
- **The first repair signal is still holding, through a week in which the Iran war widened to two new countries and the Fed delivered a rare hawkish dissent.** Price closed July 30 at \$64,706 vs \$63,557 weekly-basis (+1.8%) — still above the 200-week line despite both qualitative shocks. Still unconfirmed until a weekly close clears the 20-week average (~\$69.4K) — the real confirmation level.
- **Monthly composite -27.4/100 (Slight Bearish), 4-of-7 agree** (down from 5-of-7). Below the 20-month average (\$89,056, -27.3%), still above the 50-month (\$59,916, +8.0%); RSI 44.2; monthly MACD line +201, histogram -6,001 — deeply negative, essentially unchanged; %B 0.15 — near the lower band.
- **Quarterly composite +10.3/100 (Neutral), agreement 1-of-7** — the long tape still holds its bull structure (above the 20-quarter average, \$57,457, +12.6%) but with falling momentum (RSI 53.6, MACD histogram -1,315) and heavy quarterly OBV distribution. The weakest, noisiest layer of the stack.
- **Levels that matter:** support \$59,930, then \$57,735, then \$49,577. Resistance \$74,434, then \$81,685 (2x touch), then \$89,164; the falling weekly support trendline has re-fit sharply lower to \$48,473 as the algorithm re-weights recent price action, while the falling resistance trendline sits at \$79,926.
- ★ **June 30 remains the cycle low: \$58,526 close** — the flush tagged the Fib golden pocket (\$54,257–\$57,802 = 61.8–65% retrace of the \$15,479 → \$126,272 run) and has held a bounce to **\$64,706 (+10.6% off the low)** — through a week that brought the Iran war's widening to Iraq and Egypt and the Fed's most hawkish FOMC dissent in a decade, both of which the qualitative pillars treat as materially bearish. **\$58,526 is still the line in the sand.**



Weekly tape, scored $-28.3/100$ with 6 of 7 indicators bearish: below the 20- and 50-week averages, holding above the 200-week, broken trendline, MACD regime negative. Machine-annotated from the same data that feeds the scores.

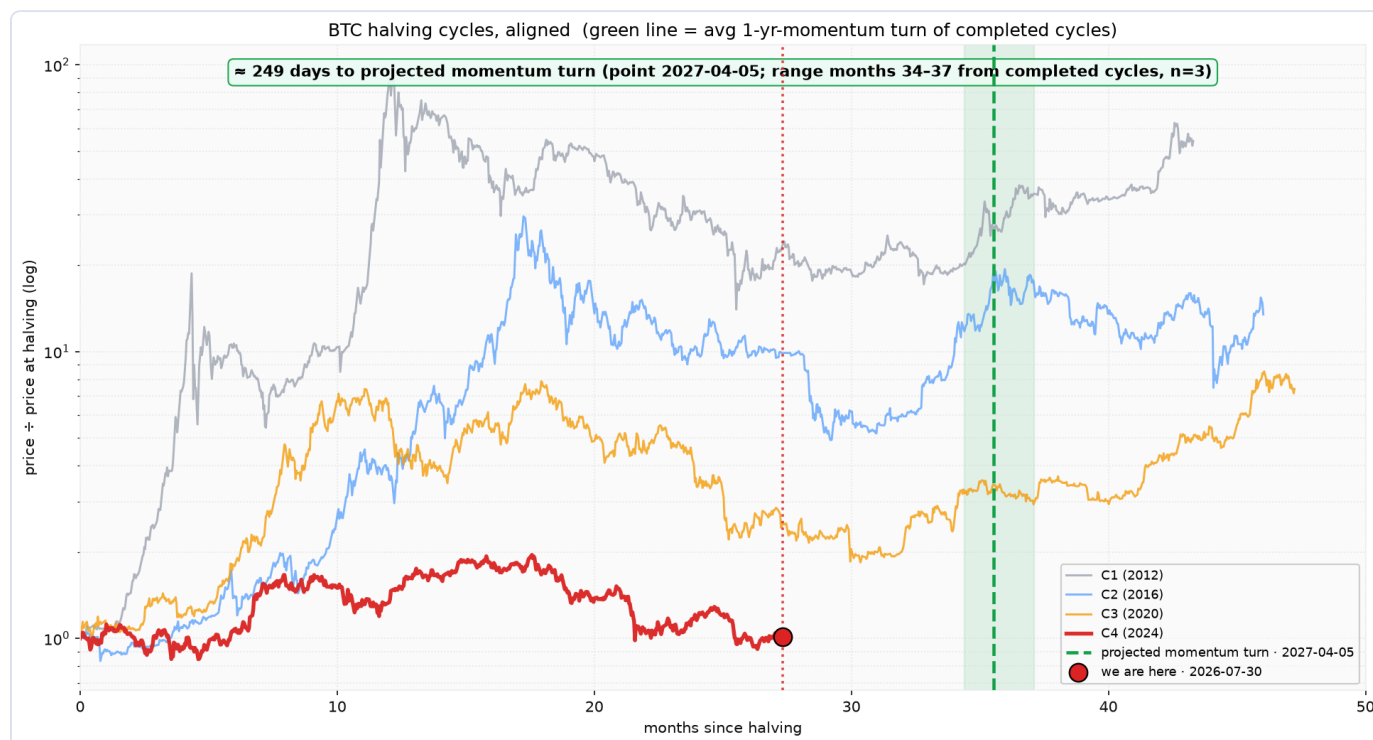
The Facts — The Cycle Clock & The Value Floor

The two slower clocks. Same data, longer lenses.

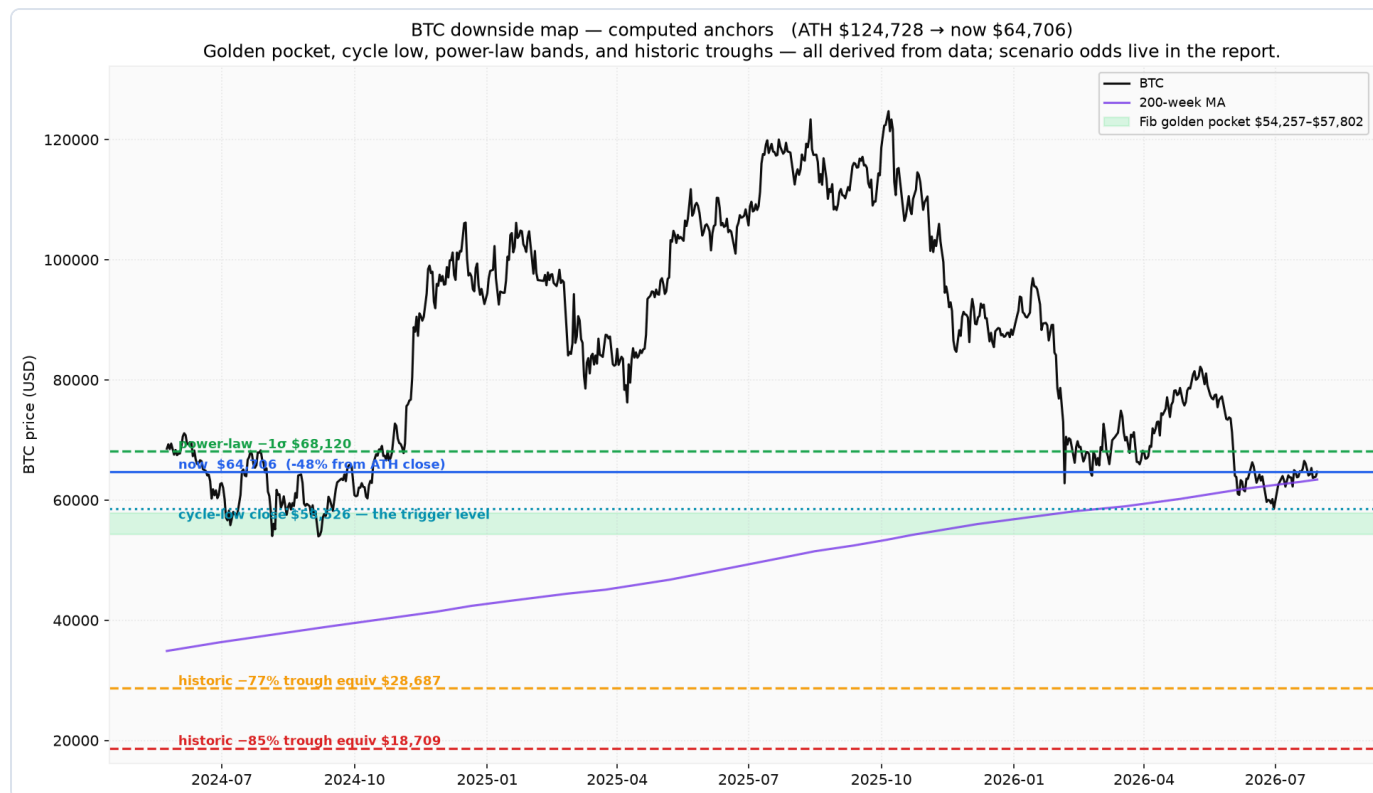
- **The halving clock says "bottoming window."** The three completed cycles topped 12–18 months after their halving (avg 15.8) and bottomed at months 26–30 (avg 28.4). This cycle topped at month 18 (Oct 6, 2025 — \$124,728) and printed its low-so-far at month 26 (June 30). **We are at month 27.33; the average-bottom date still projects to ~Aug 31, 2026.**
- ★ **The cycle analogs are the spine of the long-window scores.** Standing at this exact cycle-month in C1/C2/C3: next 1 month positive **0-of-3** (median -12.9% , weaker than last cycle's -7.1%) · next 3 months positive **0-of-3** (median -24.4% , worse still) · next 6 months positive **1-of-3** (median -15.8%) · next 12 months positive **3-of-3** (median $+24.7\%$, easing slightly from $+35.3\%$) · next 36 months positive **3-of-3** (median $+871.8\%$, up further on alignment drift). Bearish quarter, still-soft half-year, bullish year — the data's own words. *Fragility disclosure, unchanged in kind: with $n=3$, single windows keep swinging hard on pure day-count alignment as the cycle-month clock advances. Treat the analog levels as soft, the direction as the signal. And know what the 36-month number is: from month ~ 27 that window reaches month ~ 63 — it contains the NEXT halving's rally.*
- **1-year momentum turned positive at months 34–37 in every completed cycle (avg 35.5)** → projects to **~April 2027** (80% window Jan–Jul 2027) for this cycle.
- **The power-law floor:** a log-log fit over the full 16-year series (R^2 0.96) puts fair value at **\$136,438**. Fit-window honesty: starting the fit in 2011, 2012, 2013, or 2015 instead moves fair value to \$126,494 / \$131,552 / \$115,333 / \$142,386 respectively (price ~ 53 -55% below fair across that whole range), and R^2 this high is near-automatic for this kind of regression — the level is soft, but the **"below the -1σ cheap band"** verdict holds across every window tested (-1σ \$68,120; -2σ \$34,010). Every halving-era cycle trough was -77% to -85% from ATH

(\$18.7K–\$28.7K equivalent today) — the pre-halving 2011 bear went $\sim -93\%$ ($\sim \$8.7K$ equivalent), excluded from the range but in the data; this cycle is -53% so far.

- No froth in the system:** the altcoin-froth index (BTC dominance 77%, ETH/BTC 0.028) reads $z = -0.87$ — the opposite of the euphoric signature that preceded prior deep crashes. Validated on 16 years of data: high-froth days led to -39% average 12-month drawdowns vs -23% for low-froth days (gap significant, 95% CI excludes zero). *Data lags ~ 6 weeks (through May 23) — used as a tilt, not a score input.*
- ★ What we rejected:** a 5-feature logistic crash model was tested leave-one-cycle-out and **failed** (AUC 0.44 vs 0.58 for froth alone; 0.70 = useful). It is not in this report's numbers. The scores rest only on what survived out-of-sample testing.



All four halving cycles aligned (log scale, price ÷ price-at-halving). Triangles mark where 1-year momentum turned positive in each prior cycle — months 34–37, projecting to $\sim$ April 2027 for this one. The red dot is today: month 27.33, the historical bottoming window.



The downside map, updated after the June 30 flush: the projected \$54K–\$61K bottom zone was hit (\$58,526 close / \$57,735 intraday, inside the golden pocket's doorstep) and bounced. Odds re-weighted accordingly: ~40% the low is in, ~35% one more flush to \$54–\$58K, ~17% deeper bear, ~8% tail.

The Prize

Illustrative only — not price targets, not advice

From \$64,706: the 20-week average (~\$69.4K, trend-repair confirmation) is **+7%**; a retest of the \$124,728 ATH is **+93%**; the power-law fair line is **+111%**. The median 12-month analog from this cycle-month is **+24.7%** (~\$81K by mid-2027); the median 36-month analog is now +872% — **discount that one even harder than last cycle**, it leans on two regimes (2017 mania, post-COVID ZIRP) that don't repeat on demand, and it just jumped further purely on alignment drift.

The Prize doesn't feed the math — it tells you what the lean is worth acting on. Here it says: the payoff for surviving a bearish quarter is a full cycle turn.

The Other Side

The strongest honest case the model is wrong

- **n = 3.** Every long-window claim rests on three completed cycles. Three coin flips landing heads is not a law of nature.
- **This cycle already broke the pattern.** A -53% drawdown at month 26 vs -77/-85% in prior cycles could mean dampened cycles (bullish read) — or it could mean *the bottom isn't in*: a historic-depth trough from this ATH is \$18.7K–\$28.7K, far below every level in this report.
- **The ETF era changed the machine.** Prior cycles had no spot-ETF flow channel, no CME basis trade at this scale, no corporate treasuries. Analogs may simply not transfer.
- **The power law is a curve fit, not a promise.** R^2 0.96 in-sample says nothing about a structural break — a fit like this "said cheap" through most of 2022's decline too. It bounds scenarios; it doesn't time anything.

- **The tape's edge is small by construction.** The honesty cap exists because measured predictive validity is modest (froth AUC 0.58). If you removed the cap, this report would be louder — and less true.

The Triggers

Price and dated events — some prove the read right, some prove it wrong

WHEN / LEVEL	EVENT	MEANS
\$58,526	Daily close below the June 30 low	Flush scenario live — golden pocket \$54.3–57.8K is the target; re-rate 1/3-month down
~\$63,600	Weekly close above the 200-week MA (\$63,557)	Holding as of the July 30 close, despite the Iran war widening and a hawkish FOMC — still needs to hold into Sunday's weekly close
~\$69,400	Weekly close above the 20-week MA	Trend repair confirmed — re-rate 1/3-month up
~Aug 31, 2026	Historical average-bottom month (28.4) passes	A higher low after this date + improving monthly momentum = basing confirmed
Jan–Jul 2027	Projected 1-yr momentum turn window (point: ~Apr 2027)	Arriving without new lows = cycle thesis on track; re-rate 1-year up
\$50,000	Weekly close below	–60% from ATH; the dampened-cycle story is failing — see the flip line

The flip line: a decisive weekly close below **\$50,000** breaks the base case and the golden pocket both — treat the dampened-cycle thesis as broken and put the historic trough zone (\$18.7K–\$28.7K) back on the table. Above it, a bearish quarter is the plan, not a surprise.

My Decision

⊕ **MODEL DRAFT — JULY 30, 2026 · AWAITING MY STAMP**

Draft P: 39 / 37 / 46 / 67 / 70 · Quality: 0.75 / 0.70 / 0.55 / 0.60 / 0.50 · Impact: 8

This is the quantitative sleeve of the thesis — the same five windows, same formula, same table as every qualitative pillar, but computed instead of judged. The near-term read is the honest one: the tape is broken (repair signal #1 still holding through a week of widening Iran-war headlines and a hawkish FOMC surprise, still unconfirmed) and both the 1- and 3-month analogs are now 0-of-3 positive — the worst windows in the set. The 6-month window remains a near coin flip with a slight bear lean. The long read is where the quant leans bullish: cycle clock + valuation + absent froth, with the 3-year cycle-analog median jumping further bullish purely on n=3 alignment drift — flagged, not over-read.

These numbers enter the master blend as drafted. Override any P and re-stamp — the blend recomputes.

What Changed Since Last Check

Pipeline re-run July 30, 2026 — full weekly refresh (previous full run July 25, 2026)

- ★ **Price held up through a week that widened the Iran war to two new countries and delivered the Fed's most hawkish FOMC dissent in a decade: \$64,071 (Jul 25) → \$64,706 (Jul 30), +1.0%.** June 30's \$58,526 remains the cycle low; price is now +10.6% off it — a genuine, if partial, decoupling continuing this cycle.
- **Repair signal #1 is holding, not yet confirmed:** price stayed above the 200-week average (weekly-basis \$63,557) through the entire Jul 25-30 window — still unconfirmed until a weekly close clears the 20-week average (~\$69.4K). A decisive break above the falling resistance trendline (\$79,926) remains the real repair signal #2.
- **Composites moved:** weekly -41.9 → **-28.3** (agreement holding 6-of-7), monthly -29.4 → **-27.4** (agreement eased 5-of-7 → 4-of-7), quarterly +9.4 → **+10.3** (1-of-7, unchanged).
- ★ **Draft P moved 40/36/43/68/70 → 39/37/46/67/70.** Small moves across the board as cycle-month advanced from 27.17 to 27.33 and the tape composites eased slightly — no single dramatic shift this cycle, consistent with the quant pillar's mechanical, data-driven update process.
- **Data refreshed:** Bitstamp daily through July 30, 2026 (5,857 days on the master series); Coin Metrics altcoin set still through May 23 (community data lags ~6 weeks — froth is a tilt, not a score input).

Method & Sources

Everything is reproducible — the scripts and data live in the BTC-TA folder. Run order below.

- **Price data:** Bitstamp OHLC API (daily, 2011–2026, no key) via [fetch_btc_history.py](#) · Coin Metrics community reference price (2010–2011 close-only) via [build_master_history.py](#) — 5,835 days total, each row tagged by source.
- **Indicator stack:** [indicators.py](#) — SMA 20/50/200, EMA 12/26, RSI-14 (Wilder), MACD 12/26/9, Bollinger 20/2, OBV — on weekly (W-SUN), monthly, quarterly bars. [levels.py](#) — swing-cluster S/R + least-squares trendlines.
- **Scoring:** [score.py](#) — each indicator → conviction in [-100, +100] (bounded inputs mapped linearly, unbounded squashed with tanh); composite = fixed weights (MA .20, RSI .15, MACD .15, BB .15, S/R .15, OBV .10, TL .10); agreement = indicators sharing the composite's sign.
- **Cycle & value:** [cycle_analysis.py](#) — halving-aligned tops/bottoms/momentum turns, projections from C1–C3 · [downside.py](#) / [downside_map.py](#) — power-law bands (R^2 0.96) + historical troughs + 6-method bottom-zone synthesis · [test_altcoin_froth.py](#) — froth quartiles vs forward drawdowns, block-bootstrap CI.
- **The mapping (new):** [pillar_mapping.py](#) — window scores from tape composites + cycle analogs + power-law value; $P = 50 + 25 \times (S/100)$; Quality rubric per window; writes data/pillar_scores.json for the master blend.
- **Run order:** [fetch_btc_history](#) → [build_master_history](#) → [fetch_altcoin_data](#) → [indicators](#) → [levels](#) → [score](#) → [charts](#) → [cycle_analysis](#) → [downside](#) → [downside_map](#) → [pillar_mapping](#).
- **Honesty notes:** current weekly/monthly/quarterly bars are partial as of July 30 (complete-bar composites: weekly -29.2 / monthly -42.6 / quarterly +7.7) · the multi-signal crash model failed out-of-sample (AUC 0.44) and is excluded · the AUC figures are the pipeline's own attestation · froth data lags ~6 weeks · $n=3$ on all cycle claims, and analog windows can cross into the next cycle — the 6-month analog remains fragile on pure alignment drift, printed in the Facts · the 2024 halving is dated Apr 19 (US ET; block 840,000 hit Apr 20 00:09 UTC — 0.03 cycle-month effect) · S/R levels are algorithmic swing-clusters with 1–2 touches each · the bottom-odds split (40/35/17/8) is a judgment synthesis, not a model output.